

1. Hand flexion / extension ( $^{\circ}$ )
  - a. Create a device that moves with the hand while it operates the solution but stays when the hand returns. Use a protractor to measure the angle the wrist was at.
2. Duration of snack retrieval:
  - a. Time (and film) how long it takes to retrieve a product
- Snack chute:
  - Video of snack being grabbed from open space (or 3 walls if we have a box for unnecessary realism haha)
  - 1 =  $0^{\circ}$
  - 2 = xs
- Snack sack:
  - Use a regular vending machine in Chestnut to simulate opening an unrestricted flap
  - Don't reach into full depth, just approximate depth to where the product would be if snack sack was implemented
  - 1 =  $x^{\circ}$
  - 2 = xs
- Garage door:
  - Use a suitcase handle and bring from 485 - 605mm
    - That height can be relative bc suitcase is prob taller than that
  - 1 =  $x^{\circ}$
  - 2 = xs
- Sensors: to be done tomorrow
  - Doors at Myhal - watch height unless other sensors are available
  - Use [this](#) when deciding how far to stand from it
  - 1 =  $x^{\circ}$
  - 2 will use info about snack sack and add theoretical time for the flap to open (Will it? - time when someone else already opens it?)
  - 2 = xs

#### Video:

- Showing motion and functionality of recommended product (TBD, garage door?) using a low fidelity prototype

#### Snack Sack:

- Hand flexion:
  - Addison:
    - 32 degrees extension
    - 10 degrees flexion
  - Vaisnavi:
    - 15 degrees extension
    - 3 degrees flexion
  - Alina
    - 6 degrees extension
    - 15 degrees flexion
  - Helen
    - 21 degrees extension
    - 17 degrees flexion
  - Man
    - 6 degrees extension
    - 21 degrees flexion

- **Average extension: 16 degrees**
- **Average flexion: 13.2 degrees**
- Duration:
  - Addison:
    - 12.46 - 16.9 s → 4.44 s
    - 0.38 - 6.66 s → 6.28 s
  - Vaisnavi:
    - 5.42 - 8.55 s → 3.13 s
    - 0.89 - 6.25 s → 5.36 s
    - 1.18 - 5.73 s → 4.55 s
  - Alina
    - 0.00 - 4.61 s → 4.61 s
    - 0.52 - 3.92 s → 3.40 s
  - Helen
    - 2.52 - 8.67 s → 6.15 s
    - 4.97 - 9.29 s → 4.32 s
  - Man
    - 4.31 - 8.53 s → 4.22 s
    - 0.77 - 4.85 s → 4.08 s
    - 2.21 - 5.77 s → 3.56 s
  - **Average duration: 4.51 s**

Snack chute:

- Hand flexion and extension
  - Addison:
    - 5 degrees extension
    - 2 degrees flexion
  - Vaisnavi:
    - 1 degrees extension
    - 3 degrees flexion
  - Alina
    - 6 degrees extension
    - 4 degrees flexion
  - Helen
    - 6 degrees extension
    - 0 degrees flexion
  - Man
    - 8 degrees extension
    - 4 degrees flexion
  - **Average extension: 5.2 degrees**
  - **Average flexion: 2.6 degrees**
- Duration:
  - Addison:
    - 1.89
    - 3.15
  - Vaisnavi:
    - 2.35
    - 1.65
  - Alina
    - 1.23
    - 2.57
  - Helen
    - 3.59 - 5.36 s → 1.77
    - 0.00 - 2.05 s → 2.05

- Man
  - 3.25
  - 5.24
- **Average duration: 2.515 s**

Sensor:

- Addison:
  - 9 degrees extension
  - 7 degrees flexion
- Helen:
  - 6 degrees extension
  - 7 degrees flexion
- Alina:
  - 10 degrees extension
  - 8 degrees flexion
- Vaisnavi:
  - 8 degrees extension
  - 3 degrees flexion
- Man:
  - 12 degrees extension
  - 6 degrees flexion
- **Average flexion: 9 degrees**
- **Average extension: 7.2 degrees**
- Duration:
  - Helen:
    - 1.36 - 2.49 s  $\rightarrow$  1.13
    - 0.67 - 2.63 s  $\rightarrow$  1.96
  - Addison:
    - 1.43 - 3.25 s  $\rightarrow$  1.82
    - 1.85 - 3.47 s  $\rightarrow$  1.62
  - Alina:
    - 3.56 - 4.99 s  $\rightarrow$  1.43
    - 0.24 - 1.68 s  $\rightarrow$  1.44
  - Vaisnavi:
    - 1.77 - 3.54 s  $\rightarrow$  1.77
    - 0.32 - 2.23 s  $\rightarrow$  1.91
  - Man:
    - 2.11 - 3.84 s  $\rightarrow$  1.73
    - 0.00 - 1.64 s  $\rightarrow$  1.64
  - Average duration: 1.645 s
- **Total average duration: 2.515 + 1.645 = 4.16 s**

Garage door:

- Addison:
  - 4 degrees extension
  - 7 degrees flexion
- Vaisnavi:
  - 9 degrees extension
  - 5 degrees flexion
- Alina
  - 8 degrees extension
  - 6 degrees flexion
- Helen
  - 4 degrees extension

- 3 degrees flexion
- Man
  - 2 degrees extension
  - 7 degrees flexion
- **Average extension: 5.4 degrees**
- **Average flexion: 5.6 degrees**
- Duration:
  - Addison:
    - 1.32 - 2.53 s  $\rightarrow$  1.21
    - 0.94 - 2.48 s  $\rightarrow$  1.54
  - Vaisnavi:
    - 0.00 - 2.17 s  $\rightarrow$  2.17
    - 0.35 - 2.22 s  $\rightarrow$  1.87
  - Alina
    - 0.00 - 1.88 s  $\rightarrow$  1.88
    - 0.18 - 1.48 s  $\rightarrow$  1.3
  - Helen
    - 0.32 - 1.65 s  $\rightarrow$  1.33
    - 2.99 - 4.51 s  $\rightarrow$  1.52
  - Man
    - 0.23 - 1.58 s  $\rightarrow$  1.35
    - 4.25 - 6.11 s  $\rightarrow$  1.86
  - Average duration: 1.603 s

**Total duration: 1.703 + 2.515 = 4.118 s**